D. Tetragon

time limit per test: 3 seconds
memory limit per test: 256 megabytes

You're given the centers of three equal sides of a strictly convex tetragon. Your task is to restore the initial tetragon.

Input

The first input line contains one number T — amount of tests ($1 \le T \le 5 \cdot 10^4$). Each of the following T lines contains numbers $x_1, y_1, x_2, y_2, x_3, y_3$ — coordinates of different points that are the centers of three equal sides (non-negative integer numbers, not exceeding 10).

Output

For each test output two lines. If the required tetragon exists, output in the first line `YES`, in the second line — four pairs of numbers — coordinates of the polygon's vertices in clockwise or counter-clockwise order. Don't forget, please, that the tetragon should be strictly convex, i.e. no 3 of its points lie on one line. Output numbers with 9 characters after a decimal point.

If the required tetragon doen't exist, output `NO` in the first line, and leave the second line empty.

Examples

input Copy

```
3
1 1 2 2 3 3
0 1 1 0 2 2
9 3 7 9 9 8
```

output Copy

```
NO

YES
3.5 1.5 0.5 2.5 -0.5 -0.5 2.5 0.5
NO
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 23

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
371971118	Apr/21/2026 15:28	Accepted
371970820	Apr/21/2026 15:25	Wrong answer on test 1
371970681	Apr/21/2026 15:24	Wrong answer on test 1
371969823	Apr/21/2026 15:15	Wrong answer on test 2

371969656	Apr/21/2026 15:13	Wrong answer on test 2
371967030	Apr/21/2026 14:48	Wrong answer on test 2
371966312	Apr/21/2026 14:41	Wrong answer on test 2
371966012	Apr/21/2026 14:38	Wrong answer on test 1
371965854	Apr/21/2026 14:37	Compilation error
371964505	Apr/21/2026 14:24	Wrong answer on test 1

→ Problem tags

geometry math *2600

No tag edit access

→ Contest materials

- Announcement
- Tutorial

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